Controlled Actuation for Hydraulic Systems

Feedforward–Feedback Trajectory Control for Proportional Valve Architectures

A Combined Feedforward and Closed-Loop Methodology for Synchronised Flow and Pressure Control on Asymmetric Hydraulic Cylinders

Author	Sambridda Ranabhat
Organisation	Mechatronics Engineering Pvt. Ltd. (MEPL)
Location	Kathmandu, Nepal
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This document presents the governing flow and force relationships, feedforward trajectory synthesis, closed-loop compensation strategy, and process I/O architecture required to drive an asymmetric single-rod hydraulic cylinder through prescribed triangular and sinusoidal displacement profiles, using a proportional directional control valve (DCV) and pressure control valve (PCV) under PLC supervision at a 1 kHz control update rate. All valve command signals are expressed as dimensionless normalised quantities bounded to $u_{\mathrm{dcv}} \in [-1, +1]$ (full-retract to full-extend) and $u_{\mathrm{pcv}} \in [0, +1]$ (zero to full pump pressure).

Contents

1 Introduction and Scope

This proposal defines the control methodology required to drive a single-rod hydraulic cylinder through prescribed displacement-time trajectories—specifically triangular and sinusoidal waveforms—using independently actuated flow and pressure control elements under programmable logic controller (PLC) supervision.

The system architecture comprises a proportional directional control valve (DCV) governing flow rate and therefore piston velocity, a pressure control valve (PCV) governing system pressure and therefore actuation force, and a PLC executing a closed-loop control routine at a 1 kHz update rate. The cylinder is treated as an asymmetric actuator in which the cap-end area exceeds the rod-end area, requiring distinct flow demands for extension and retraction at equal target velocities.

All valve command signals produced by the PLC are **dimensionless normalised quantities**: the DCV command $u_{\mathrm{dcv}} \in [-1, +1]$ represents the proportion of the valve's maximum rated flow, with negative values commanding retraction and positive values commanding extension; the PCV command $u_{\mathrm{pcv}} \in [0, +1]$ represents the proportion of the pump's maximum rated pressure. Both signals map linearly onto the physical analog output channel (0–10 V or 4–20 mA).

The proposal addresses two architectural configurations for the PCV and derives full command-signal waveforms for each:

- **Configuration A** — PCV is a conventional fixed-setting relief valve; trajectory shaping is performed entirely by the DCV, and u_{pcv} is held constant throughout operation.
- **Configuration B** — PCV is a proportional valve accepting a dynamic PLC command; both u_{dcv} and u_{pcv} are modulated in real time to match the instantaneous kinematic and force demand.

2 System Variables and Physical Foundations

2.1 Symbol Table

The control strategy is developed from the variable set summarised in Table 1. All quantities are functions of time unless otherwise stated.

Table 1: System variables and physical parameters

Symbol	Description
$S(t)$, $v(t)$, $a(t)$	Target piston position, velocity, and acceleration
$S_{\text{act}}(t)$	Measured piston position from primary and backup sensors
$u_{\text{dcv}}(t)$	Normalised DCV command $\in [-1, +1]$; negative = retraction, positive = extension, zero = valve closed at centre
$u_{\text{pcv}}(t)$	Normalised PCV command $\in [0, +1]$; zero = minimum pilot pressure, unity = maximum rated pump pressure P_{pump}
A_{cap}, A_{rod}	Cap-end and rod-end piston areas [m^2]; $A_{\text{cap}} > A_{\text{rod}}$ for any single-rod cylinder
Q_n, ΔP_n	Rated volumetric flow [m^3/s] and rated pressure drop [Pa] of the DCV at full opening; from manufacturer datasheet
P_{pump}	Maximum rated pump delivery pressure [Pa]
$P_{\text{sys}}(t)$	Instantaneous system pressure, regulated by the PCV
$P_{\text{cyl}}(t)$	Back-pressure in the passive (exhaust) cylinder chamber
P_{tank}	Return-line back-pressure ($\approx$ 0.5–1 MPa typical)
P_{relief}	Fixed relief valve set-point (Configuration A only)
μ	Dynamic viscosity of VG46 hydraulic fluid; temperature-dependent over 30–55 °C operating range
m, $B(\mu)$	Total moving mass (piston + load) [kg] and viscous damping coefficient [$\text{N} \cdot \text{s/m}$]
F_{ext}	Constant external load force opposing the stroke direction [N]
β_e	Effective bulk modulus of the hydraulic fluid, accounting for dissolved air and hose compliance [Pa]; typically $0.7\text{–}1.0 \times 10^9$ Pa for VG46 with circuit compliance
V_t	Total trapped fluid volume (cylinder + hose + manifold) at mid-stroke [m^3]
K_v	Valve flow coefficient $= Q_n/\sqrt{\Delta P_n}$ [m^3/s/$\sqrt{\text{Pa}}$]

2.2 Orifice Flow Relationship

The volumetric flow through the proportional DCV metering edge (Huade 4WRZE or equivalent) obeys the hydraulic orifice equation derived from Torricelli's theorem applied to a sharp-edged restriction:[1]

$$Q = K_v \cdot u_{\text{dcv}} \cdot \sqrt{\Delta P} \tag{1}$$

Term definitions:

- Q — volumetric flow rate through the valve [m^3/s]. The sign of u_{dcv} carries the direction: $Q > 0$ denotes flow into the cap-end (extension) and $Q < 0$ denotes flow out of the cap-end (retraction).
- $K_v = Q_n/\sqrt{\Delta P_n}$ — the valve flow coefficient [m^3/s/$\sqrt{\text{Pa}}$]; a manufacturer-supplied constant that encapsulates the valve's discharge coefficient, orifice area, and fluid

[1] Merritt, H.E. (1967), *Hydraulic Control Systems*, Wiley, §3.2 ("The Hydraulic Orifice"); see also ISO 4411:2008, *Hydraulic fluid power — Determination of pressure differential–flow characteristics for valves*, and ISO 10770-1:2009 for proportional valve performance characterisation.

density at the rated operating condition. For the Huade 4WRZE, K_v must be read from the hydraulic datasheet for the specific nominal flow variant installed.

- $u_{\text{dcv}} \in [-1, +1]$ — dimensionless normalised spool displacement. At $u_{\text{dcv}} = +1$ the spool is fully open in the extension direction; at -1 it is fully open for retraction; at 0 the spool centres and flow is blocked. A 10 V output maps to $u_{\text{dcv}} = +1$, 5 V maps to 0 (centre), and 0 V maps to -1. For a 4–20 mA interface: 20 mA $\leftrightarrow +1$, 12 mA $\leftrightarrow 0$, 4 mA $\leftrightarrow -1$.
- $\Delta P = P_{\text{sys}} - P_{\text{cyl}}$ — pressure differential across the valve metering land [Pa]; the available hydraulic driving head is reduced by any back-pressure in the active cylinder chamber.

2.3 Normalised Command Signal Derivation

Rearranging Equation 1 to express the required DCV command as a function of the flow demand $Q = A \cdot v(t)$ and the prevailing pressure differential:

$$u_{\text{dcv}} = \frac{Q}{Q_n}\sqrt{\frac{\Delta P_n}{\Delta P}} = \frac{A \cdot v(t)}{Q_n}\sqrt{\frac{\Delta P_n}{P_{\text{sys}} - P_{\text{cyl}}}} \tag{2}$$

This is the *general feedforward DCV expression*. It yields a dimensionless value in $[-1, +1]$ provided the required flow does not exceed the valve's rated capacity. If $|u_{\text{dcv}}|$ exceeds unity, the requested velocity cannot be achieved at the prevailing pressure differential and the trajectory parameters or valve specification must be revised.

The normalised PCV command is the ratio of the required system pressure to the pump's maximum delivery:

$$u_{\text{pcv}} = \frac{P_{\text{sys}}(t)}{P_{\text{pump}}} \in [0, 1] \tag{3}$$

Equation 1 also reveals the coupling between the two control channels: the DCV command cannot be evaluated independently of P_{sys}, since the achievable flow depends on the square root of the pressure differential supplied by the PCV. This is why the DCV and PCV commands must be computed together each control cycle rather than independently.

3 Pressure Control Strategy: Architectural Options

3.1 Configuration A — PCV as a Fixed Safety Limit

In this configuration the PCV is a conventional relief valve that is not actively modulated during normal operation. It is mechanically pre-set to a fixed system pressure P_{relief}, and the corresponding PLC output is held constant throughout the trajectory:

$$u_{\text{pcv}} = \frac{P_{\text{relief}}}{P_{\text{pump}}} = \text{const} \tag{4}$$

Term definitions:

- P_{relief} — the mechanically set relief pressure [Pa]; must be chosen high enough to supply the peak force demand of the most demanding instant of the trajectory (direction reversal for triangular, maximum deceleration for sinusoidal). Undersetting this value will cause the piston to stall.
- P_{pump} — normalising denominator; the pump's maximum rated delivery pressure [Pa].

Under Configuration A, $P_{\text{sys}} \approx P_{\text{relief}} = \text{const}$. This constant pressure makes the denominator of Equation 2 a fixed number evaluated once at commissioning, greatly simplifying the DCV computation. The amplitude scaling factors for the two stroke directions become:

Configuration A — DCV amplitude coefficients (evaluated at commissioning):

$$k_{\text{ext}} = \frac{A_{\text{cap}}}{Q_n}\sqrt{\frac{\Delta P_n}{P_{\text{relief}} - P_{\text{cyl,ss}}}} \tag{5}$$

$$k_{\text{ret}} = \frac{A_{\text{rod}}}{Q_n}\sqrt{\frac{\Delta P_n}{P_{\text{cyl,ss}} - P_{\text{tank}}}} \tag{6}$$

where $P_{\text{cyl,ss}}$ is the steady-state back-pressure in the passive cylinder chamber under load, measured or estimated during initial commissioning. These constants are substituted into the trajectory equations in Sections 4 and 5.

The computational advantage of Configuration A comes at an energy cost: oil is pumped at full relief pressure throughout the cycle, including the low-force phases where only a fraction of that pressure is needed. The excess energy is dissipated as heat across the relief valve.

3.2 Configuration B — Active Proportional Pressure Control

If the installed PCV accepts a dynamic analogue command, system pressure is actively modulated to match the instantaneous force demand. This eliminates the continuous energy loss of Configuration A and enables tighter force control, but requires a real-time floating-point force calculation and a square-root evaluation every control cycle.

Architectural decision required. The dynamic $u_{\text{pcv}}(t)$ equations developed in Sections 4 and 5 assume Configuration B. If the installed PCV is a fixed-setting relief valve (Configuration A), those equations reduce to Equation 4 and the DCV amplitude factors become the fixed constants k_{ext} and k_{ret}. The installed PCV type must be confirmed against the hardware specification before control software is finalised.

4 Trajectory Synthesis: Triangular Displacement Profile

A triangular displacement profile requires constant velocity during both extension and retraction strokes, with an abrupt direction reversal at each peak. Figure 1 shows the target displacement alongside the normalised DCV and PCV command waveforms for both configurations.

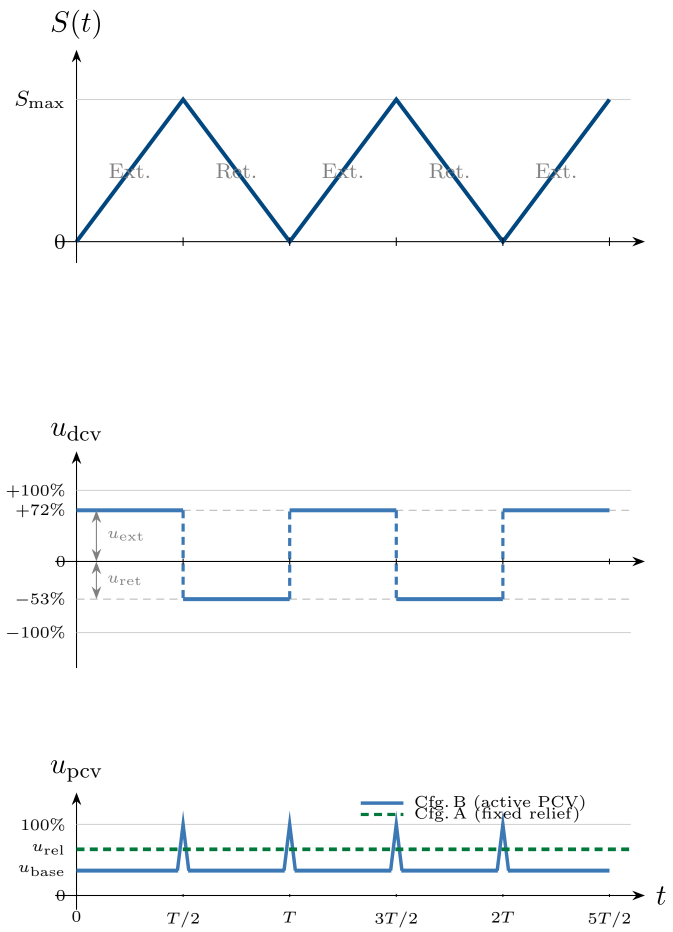

Figure 1: Three-panel command signal overview for the triangular displacement profile. **Top**: target displacement $S(t)$; constant-velocity extension and retraction strokes alternate with instantaneous direction reversals at each peak. **Middle**: normalised DCV command $u_{\text{dcv}} \in [-1, +1]$; the waveform is an asymmetric square wave. The extension opening $u_{\text{ext}} = 72\%$ exceeds the retraction opening $u_{\text{ret}} = 53\%$ (illustrative values) because the larger cap-end bore requires proportionally more flow to achieve the same piston velocity as the smaller rod-end bore. Vertical dashed segments show the instantaneous direction reversal. **Bottom**: normalised PCV command $u_{\text{pcv}} \in [0, +1]$; Configuration B (blue) holds a low steady-state baseline u_{base} covering friction and external load, with momentary pressure spikes to 100% at each reversal point to supply the inertial impulse needed to arrest and reverse piston momentum; Configuration A (green dashed) maintains a fixed relief setting u_{rel} throughout, which must be set above u_{base} but results in continuous energy dissipation across the relief valve.

4.1 DCV Flow Command

Because the target velocity is constant during each stroke segment, the required flow is likewise constant, making the DCV command a block (square-wave) profile. Applying Equation 2 with $Q = A \cdot v_0$:[2]

[2]Continuity equation applied to a rigid piston: $Q = A\,v$; see Merritt (1967), §2.1. Area asymmetry ($A_{\text{cap}} \neq A_{\text{rod}}$) is the defining geometric characteristic of a single-rod cylinder.

Configuration B (P_{sys} is time-varying and must be evaluated each control cycle):

$$u_{\mathrm{dcv}} = +\frac{A_{\mathrm{cap}}\, v_0}{Q_n}\sqrt{\frac{\Delta P_n}{P_{\mathrm{sys}} - P_{\mathrm{cyl}}}} \quad (v = +v_0,\ \text{extension}) \tag{7}$$

$$u_{\mathrm{dcv}} = -\frac{A_{\mathrm{rod}}\, v_0}{Q_n}\sqrt{\frac{\Delta P_n}{P_{\mathrm{cyl}} - P_{\mathrm{tank}}}} \quad (v = -v_0,\ \text{retraction}) \tag{8}$$

Configuration A ($P_{\mathrm{sys}} = P_{\mathrm{relief}} = \mathrm{const} \Rightarrow$ constant command amplitudes):

$$u_{\mathrm{dcv}} = +k_{\mathrm{ext}} \quad = \mathrm{const} \quad (\text{extension}) \tag{9}$$

$$u_{\mathrm{dcv}} = -k_{\mathrm{ret}} \quad = \mathrm{const} \quad (\text{retraction}) \tag{10}$$

Term definitions (Eqs. 7–10):

- v_0 — constant stroke velocity magnitude [m/s]; the same in extension and retraction by definition of the triangular profile.
- $A_{\mathrm{cap}}, A_{\mathrm{rod}}$ — bore areas [m^2]; the ratio $A_{\mathrm{cap}}/A_{\mathrm{rod}} > 1$ is why $|u_{\mathrm{ext}}| > |u_{\mathrm{ret}}|$ at the same velocity.
- P_{cyl} — exhaust-side back-pressure in the passive cylinder chamber; equals approximately P_{tank} under ideal conditions but may be elevated by fluid inertia and flow losses in practice.
- $k_{\mathrm{ext}}, k_{\mathrm{ret}}$ — fixed amplitude coefficients defined in Equations 5–6 and evaluated once at commissioning. Under Configuration A, both are constants for the full duration of operation.

The resulting DCV output is an asymmetric square wave: $|u_{\mathrm{ext}}| > |u_{\mathrm{ret}}|$ because filling the larger cap-end requires a wider valve opening than evacuating the smaller rod-end at equal piston speed.

4.2 PCV Pressure Command: Configuration B

Acceleration is zero along each constant-velocity segment; the only force demand is viscous friction and external load. Applying Newton's second law ($F = ma + B(\mu)v + F_{\mathrm{ext}}$) with $a = 0$:[3]

$$u_{\mathrm{pcv}} = \frac{F_{\mathrm{ext}} + B(\mu)\, v_0}{A_{\mathrm{cap}}\, P_{\mathrm{pump}}} + \frac{\Delta P_{\mathrm{valve,min}}}{P_{\mathrm{pump}}} \tag{11}$$

Term definitions:

- F_{ext} — constant external load opposing the stroke [N].
- $B(\mu)\, v_0$ — viscous drag force [N]; $B(\mu)$ varies with fluid temperature through the viscosity μ of VG46.
- $A_{\mathrm{cap}}\, P_{\mathrm{pump}}$ — the denominator normalises the force [N] into the $[0, 1]$ command range.

[3] Newton's second law for a hydraulic piston-and-load rigid body; Jelali, M. & Kroll, A. (2004), *Hydraulic Servo-systems: Modelling, Identification and Robust Control*, Springer, §2.3.

- $\Delta P_{\text{valve,min}}/P_{\text{pump}}$ — minimum pressure margin (typically 0.5–1 MPa) ensuring the spool remains in its linear metering region throughout the stroke.

At each triangular peak, the target acceleration is instantaneous (a velocity step), requiring a momentary pressure spike $u_{\text{pcv}} \to 1$ to supply the inertial impulse needed to arrest and reverse piston momentum without lag. Outside these brief reversal intervals, u_{pcv} returns to the low steady-state value of Equation 11, as shown in the lower panel of Figure 1.

4.3 PCV Pressure Command: Configuration A

Under Configuration A, the PLC output to the PCV is the constant of Equation 4 throughout the cycle. The relief valve opens transiently at each direction reversal to dump excess flow while the DCV arrests the piston; the PLC takes no action on the pressure side. This is the flat green line in Figure 1: u_{pcv} is invariant regardless of the instantaneous load or velocity.

5 Trajectory Synthesis: Sinusoidal Displacement Profile

A sinusoidal displacement profile requires continuous, smoothly varying acceleration and deceleration throughout the stroke. The target kinematics are:[4]

$$S(t) = X_0 \sin(\omega t) \tag{12}$$
$$v(t) = X_0 \omega \cos(\omega t) \tag{13}$$
$$a(t) = -X_0 \omega^2 \sin(\omega t) \tag{14}$$

Figure 2 shows the target displacement alongside the normalised command waveforms. The 90° phase offset between displacement and velocity is fundamental: the valve is widest open when the piston is at mid-stroke moving fastest ($v = X_0\omega$, $S = 0$), and fully closed when the piston is at the stroke extreme ($v = 0$, $S = \pm X_0$).

[4]Standard kinematic chain for simple harmonic motion; the derivatives follow directly from calculus. $\omega = 2\pi f$ is the angular frequency [rad/s].

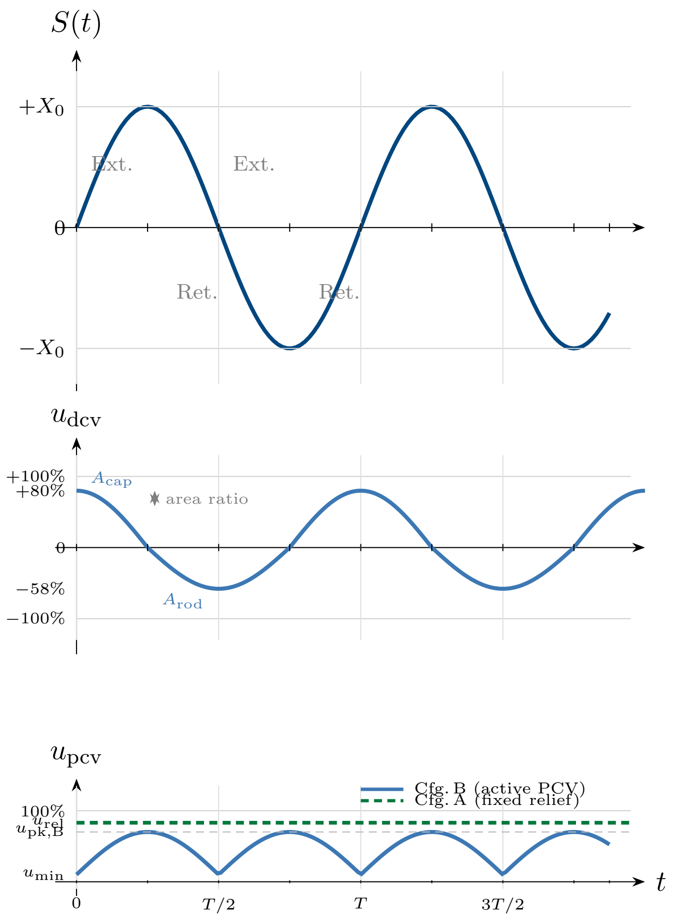

Figure 2: Three-panel command signal overview for the sinusoidal displacement profile. **Top**: target displacement $S(t) = X_0 \sin(\omega t)$; vertical grey guides mark the zero-crossing instants where velocity is maximum and force demand from inertia is zero. **Middle**: normalised DCV command u_{dcv}; the waveform tracks $\cos(\omega t)$ but with asymmetric peak amplitudes in the extension and retraction half-cycles (+80% vs. −58%, illustrative values reflecting $A_{\text{cap}}/A_{\text{rod}}$). The small kink at each zero crossing ($t = T/2, T, \ldots$) is a physical derivative discontinuity caused by the switch in area scaling from A_{cap} to A_{rod}; both amplitudes pass through zero at the same instant, so the command itself is continuous. **Bottom**: normalised PCV command $u_{\text{pcv}} \in [0, +1]$; Configuration B (blue) oscillates at *twice* the motion frequency ω, peaking when the piston is at maximum displacement (maximum deceleration force) and reaching its minimum when the piston passes through zero (maximum velocity but zero inertial force); Configuration A (green dashed) must be set to $u_{\text{rel}} > u_{\text{pk,B}}$ to supply adequate force at the worst-case instant, wasting pressure throughout the remainder of the cycle.

5.1 DCV Flow Command

A sinusoidal displacement requires a cosine velocity profile that directly determines the DCV command. Applying Equation 2 with $Q = A\,v(t) = A\,X_0\omega\cos(\omega t)$:[5]

[5]Merritt (1967), §3.3; the cosine velocity derived from the sinusoidal position setpoint is the direct feedforward flow demand.

Configuration B ($P_{\mathrm{sys}}(t)$ time-varying):

$$u_{\mathrm{dcv}} = \frac{A_{\mathrm{cap}}\, X_0 \omega \cos(\omega t)}{Q_n} \sqrt{\frac{\Delta P_n}{P_{\mathrm{sys}}(t) - P_{\mathrm{cyl}}(t)}} \qquad \big[\cos(\omega t) > 0, \text{extending}\big] \tag{15}$$

$$u_{\mathrm{dcv}} = \frac{A_{\mathrm{rod}}\, X_0 \omega \cos(\omega t)}{Q_n} \sqrt{\frac{\Delta P_n}{P_{\mathrm{cyl}}(t) - P_{\mathrm{tank}}}} \qquad \big[\cos(\omega t) < 0, \text{retracting}\big] \tag{16}$$

Configuration A ($P_{\mathrm{sys}} = P_{\mathrm{relief}} = \text{const} \Rightarrow$ time-invariant amplitude factors):

$$u_{\mathrm{dcv}} = K_{\mathrm{ext}} \cdot \cos(\omega t) \qquad \big[\cos(\omega t) > 0\big] \tag{17}$$

$$u_{\mathrm{dcv}} = K_{\mathrm{ret}} \cdot \cos(\omega t) \qquad \big[\cos(\omega t) < 0\big] \tag{18}$$

where the constants are defined as:

$$K_{\mathrm{ext}} = \frac{A_{\mathrm{cap}}\, X_0 \omega}{Q_n} \sqrt{\frac{\Delta P_n}{P_{\mathrm{relief}} - P_{\mathrm{cyl,ss}}}}, \qquad K_{\mathrm{ret}} = \frac{A_{\mathrm{rod}}\, X_0 \omega}{Q_n} \sqrt{\frac{\Delta P_n}{P_{\mathrm{cyl,ss}} - P_{\mathrm{tank}}}}$$

Term definitions:

- X_0 — half-amplitude of the sinusoidal stroke [m]; half the total travel.
- ω — angular frequency [rad/s] $= 2\pi f$.
- $\cos(\omega t)$ — the normalised velocity waveform, 90° ahead of the displacement; this is the feedforward flow demand shape.
- **Sign convention for retraction**: when $\cos(\omega t) < 0$ (retracting), $K_{\mathrm{ret}} > 0$ and the cosine value is negative, so Equations 16 and 18 automatically produce $u_{\mathrm{dcv}} < 0$ (retraction command) without requiring an explicit sign change.
- **Configuration A advantage**: because K_{ext} and K_{ret} are constants, the DCV command requires only a single multiplication each control cycle (multiply by $\cos(\omega t)$). Configuration B requires a floating-point square-root evaluation every millisecond, which must be benchmarked against the selected PLC's floating-point instruction throughput.

5.2 PCV Pressure Command: Configuration B

The force demand varies continuously because the mass is constantly accelerating or decelerating. Applying Newton's second law in full:[6]

$$F_{\mathrm{total}}(t) = -m\, X_0 \omega^2 \sin(\omega t) + B(\mu)\, X_0 \omega \cos(\omega t) + F_{\mathrm{ext}} \tag{19}$$

$$u_{\mathrm{pcv}}(t) = \frac{\big|F_{\mathrm{total}}(t)\big|}{A_{\mathrm{cap}}\, P_{\mathrm{pump}}} + \frac{\Delta P_{\mathrm{valve,min}}}{P_{\mathrm{pump}}} \tag{20}$$

Term definitions:

[6] Merritt (1967), §3.4. The combined inertia and viscous damping force is the standard equation of motion for a hydraulic servo actuator under sinusoidal excitation.

- $-m\,X_0\omega^2\sin(\omega t)$ — **inertial force term** [N]; proportional to mass and to the *square* of the operating frequency. This term is in phase with $-S(t)$: force demand is maximum at maximum displacement (maximum deceleration) and zero at the zero crossing (no acceleration). Because it scales with ω^2, the inertial force grows rapidly with frequency and becomes the dominant constraint at high operating frequencies (see Section 7).
- $B(\mu)\,X_0\omega\cos(\omega t)$ — **viscous damping force term** [N]; proportional to velocity, it is 90° ahead of displacement and exactly in phase with the DCV command. It scales only linearly with ω.
- F_{ext} — constant external load [N], elevating the pressure baseline by a constant offset.
- $|F_{\text{total}}|$ — the absolute value is taken because the PCV controls pressure magnitude; the DCV direction signal determines which cylinder chamber receives the high-pressure supply.

Because $F_{\text{total}}(t) = R\sin(\omega t + \varphi) + F_{\text{ext}}$ (a single sinusoid at frequency ω phase-shifted by $\varphi = \arctan\big(B(\mu)/m\omega\big)$), the quantity $|F_{\text{total}}(t)|$ oscillates at *twice* the motion frequency when $F_{\text{ext}} < R$. This double-frequency pressure demand is visible in the lower panel of Figure 2: the Config B curve shows two pressure peaks per displacement cycle, corresponding to the two instants per cycle at which piston deceleration (and therefore required braking force) is greatest.

5.3 PCV Pressure Command: Configuration A

The PCV is held constant at $u_{\text{pcv}} = P_{\text{relief}}/P_{\text{pump}}$ (Equation 4). This flat line must be set above the peak Configuration B demand $u_{\text{pk,B}}$ (see Figure 2) to guarantee that sufficient hydraulic force is available at the worst-case deceleration point. If the relief setting is too low, the piston will fail to decelerate on schedule and will overshoot the stroke end.

6 Closed-Loop Feedback Compensation

The feedforward expressions in Sections 4 and 5 account for the full theoretical motion demand but do not compensate for oil compressibility, viscosity drift with temperature, valve deadband, or unmodelled load disturbances. A cascaded PID correction layer is applied to the DCV command to close this gap.

6.1 Control Structure

The position error is defined as:

$$e(t) = S_{\text{target}}(t) - S_{\text{act}}(t) \tag{21}$$

The total normalised DCV command, clamped to $[-1, +1]$ before output, is:[7]

[7] Åström, K.J. & Hägglund, T. (1995), *PID Controllers: Theory, Design and Tuning*, 2nd ed., ISA; Jelali & Kroll (2004), §5.3.

$$u_{\text{dcv,total}}(t) = \underbrace{u_{\text{ff}}(t)}_{\text{Secs. 4–5}} + \underbrace{K_p\, e(t)}_{\text{proportional}} + \underbrace{K_i \int_0^t e(\tau)\,\mathrm{d}\tau}_{\text{integral}} + \underbrace{K_d \frac{\mathrm{d}e}{\mathrm{d}t}}_{\text{derivative}} \tag{22}$$

Term definitions:

- $u_{\text{ff}}(t)$ — feedforward term from Sections 4 or 5; provides the bulk of the valve command from the trajectory setpoint alone, so the PID terms correct only the residual tracking error.
- $K_p\, e(t)$ — proportional correction; increases or decreases valve opening in direct proportion to the instantaneous position lag. High K_p improves tracking but introduces oscillation.
- $K_i \int e\,\mathrm{d}t$ — integral correction; accumulates residual steady-state error caused by friction or deadband and drives it to zero over time. Prone to windup at valve saturation; see Section 8.2.
- $K_d\, \mathrm{d}e/\mathrm{d}t$ — derivative correction; anticipates the rate of change of error, providing braking action at trajectory peaks and damping oscillation introduced by high K_p.
- The total $u_{\text{dcv,total}}$ must be **clamped** to $[-1, +1]$ before being written to the analog output; a computed value outside this range indicates that the combined feedforward plus PID demand exceeds the valve's rated capacity.

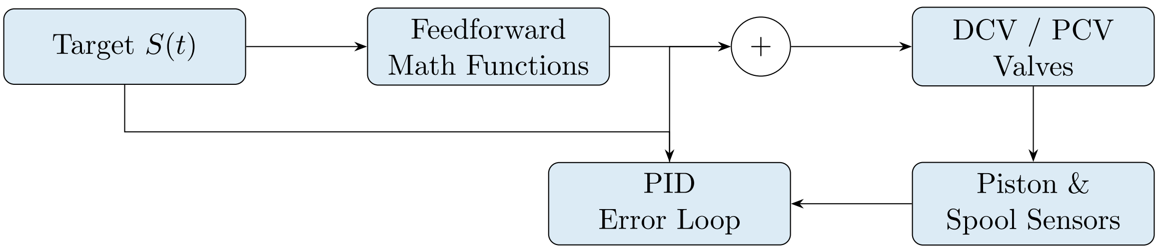

Figure 3: Cascaded feedforward and feedback control architecture

6.2 Cascade Structure

- **Outer loop (piston position).** The position sensor feeds $S_{\text{act}}(t)$ to the PID, which adds a correction to the feedforward DCV output wherever the piston lags the setpoint.
- **Inner loop (spool position).** The DCV's built-in spool-position servo ensures the valve drives its spool to exactly the commanded position, eliminating drift from temperature-induced changes in pilot-oil viscosity.

6.3 Tuning Procedure

1. Set $K_p = K_i = K_d = 0$; run the feedforward trajectory and record the position lag.
2. Increase K_p until the lag closes, accepting minor oscillation or chatter.
3. Increase K_d to damp the oscillation and limit overshoot at trajectory peaks.
4. Introduce a small K_i only after K_p and K_d are stable; verify anti-windup behaviour at the turning points before releasing for continuous operation.

Integral windup. If the valve saturates ($|u_{\text{dcv,total}}| = 1$) while the piston has not yet reached the target, the integral term continues to accumulate. When the trajectory reverses direction, this accumulated integral slows the controller's response and risks driving the piston into its mechanical end stop. Anti-windup logic (Section 8.2) must be implemented before any continuous operation.

7 Operational Frequency Range

The maximum trajectory frequency that the system can track reliably is bounded by four independent constraints. Each is derived below, and Table 2 summarises which are mass-dependent.

7.1 Hydraulic Natural Frequency

The linearised hydraulic actuator has an undamped natural frequency of:[8]

$$\omega_h = A_{\text{cap}} \sqrt{\frac{4\beta_e}{m\,V_t}} \qquad \Longrightarrow \qquad f_h = \frac{\omega_h}{2\pi} \tag{23}$$

Term definitions:

- β_e — effective bulk modulus [Pa]; for VG46 with circuit compliance and dissolved air, $\beta_e \approx 0.7\text{–}1.0 \times 10^9$ Pa, substantially below the pure-oil value. A lower β_e (more compliant circuit) reduces ω_h and therefore the achievable tracking bandwidth.
- V_t — total trapped fluid volume [m^3] at mid-stroke; includes cylinder bore, hose, and manifold on both sides of the piston.
- m — total moving mass [kg] including piston, rod, and payload.
- A_{cap} — cap-end bore area [m^2].

Hydraulic actuators are lightly damped ($\zeta \approx 0.1$–0.3). To maintain adequate phase margin, the operating frequency must be well below f_h:

$$f_{\text{op}} < \frac{f_h}{3} \tag{24}$$

Mass dependence: $\omega_h \propto 1/\sqrt{m}$; doubling the load mass reduces the hydraulic natural frequency by $\sqrt{2}$, tightening Equation 24 by the same factor.

7.2 Flow-Limited Maximum Frequency

For sinusoidal motion, peak flow occurs at the zero crossing of $S(t)$ where $|\cos(\omega t)| = 1$:

$$Q_{\text{peak}} = A_{\text{cap}}\,X_0\,\omega \tag{25}$$

Setting $Q_{\text{peak}} \le Q_{\text{pump}}$ and solving for f:

[8] Merritt (1967), eq. (3.26); Manring, N.D. (2005), *Hydraulic Control Systems*, Wiley, §7.2. This is the hydraulic resonance frequency of the piston-mass-oil-column system.

$$f_Q = \frac{Q_{\text{pump}}}{2\pi\, A_{\text{cap}}\, X_0} \tag{26}$$

Mass-independent. The flow limit depends only on pump capacity, bore area, and stroke amplitude. It can be relaxed by increasing Q_{pump} or reducing X_0.

7.3 Force-Limited Maximum Frequency

Peak inertial force occurs at maximum displacement where $|a| = X_0\omega^2$:

$$F_{\text{inertia,peak}} = m\, X_0\, \omega^2 \tag{27}$$

Setting $F_{\text{inertia,peak}} \le P_{\text{pump}}\, A_{\text{cap}}$:

$$f_F = \frac{1}{2\pi}\sqrt{\frac{P_{\text{pump}}\, A_{\text{cap}}}{m\, X_0}} \tag{28}$$

Mass-dependent: $f_F \propto 1/\sqrt{m}$. This is typically the binding constraint for heavy loads: a payload ten times heavier reduces the force-limited frequency by $\sqrt{10} \approx 3.16$.

7.4 Valve Bandwidth Constraint

The DCV spool-positioning servo has its own frequency response. For the Huade 4WRZE or equivalent proportional DCV, the spool −3 dB bandwidth is typically 50–100 Hz. Commands above the valve's bandwidth are attenuated before they reach the hydraulic circuit, degrading tracking fidelity. A practical guideline for trajectory following is:

$$f_{\text{op}} < \frac{f_{\text{valve}}}{5} \approx 10\text{–}20\,\text{Hz} \tag{29}$$

This constraint is **independent of mass** and is set by the valve model alone. For most medium-duty proportional DCVs, it is the practical binding limit.

7.5 Summary: Maximum Achievable Frequency

$$f_{\max} = \min\left(\frac{f_h}{3},\ \ f_Q,\ \ f_F,\ \ \frac{f_{\text{valve}}}{5}\right) \tag{30}$$

Table 2: Frequency constraint summary

Constraint	**Eq.**	**Mass-dep.**	**Typically binding when**
Hydraulic resonance $f_h/3$	24	Yes ($\propto 1/\sqrt{m}$)	Heavy loads; large-bore or long cylinders with compliant hose
Flow (pump) limit f_Q	26	No	High frequency, small bore, long stroke, limited pump flow
Force (inertia) limit f_F	28	Yes ($\propto 1/\sqrt{m}$)	Heavy payloads at moderate-to-high frequency
Valve bandwidth $f_{\text{valve}}/5$	29	No	Almost all proportional DCV applications; often the dominant limit below ~ 20 Hz

In summary: the system can operate at significantly higher frequencies under light loads (both f_h and f_F improve as $1/\sqrt{m}$), but the valve bandwidth imposes a hard ceiling that is independent of load. For a typical medium-duty installation (bore 50–80 mm, load 100–500 kg), the practical upper limit is $f_{\max} \approx 10$–20 Hz, controlled by the valve bandwidth unless the load is very light.

8 PLC Implementation Considerations

8.1 Control Loop Rate

The PID calculation must execute at the same 1 kHz update rate as the data-acquisition cycle. In practice, a true 1 kHz closed loop requires the complete cycle—read two analog inputs, evaluate feedforward plus PID, write two analog outputs—to complete within 1 ms. Standard analog modules convert in 1–5 ms per channel and therefore limit the achievable closed-loop rate to 200–500 Hz. Given that the Huade 4WRZE spool bandwidth is ≈ 50–100 Hz, a 500 Hz PLC update rate already provides a $\times 5$–$\times 10$ oversampling ratio relative to the valve's own dynamics, which is adequate for stable closed-loop control without requiring specialist ultra-fast hardware. The 1 kHz specification should be reviewed against the actually required trajectory frequency before committing to a hardware platform.

8.2 Anti-Windup Logic

When $|u_{\text{dcv,total}}|$ saturates at 1 and the piston has not yet reached the target position, integral accumulation must be halted. Without this, the controller becomes slow to reverse the integral term at trajectory turning points, risking overshoot into mechanical end stops.

8.3 Deadband Compensation

Proportional valves exhibit a deadband around the centred spool position within which no flow occurs. A small offset must be added to the PID output as soon as the command moves off-centre, so the spool clears the deadband immediately and avoids dead-time in the motion response.

8.4 Commissioning Sequence

1. Set $K_p = K_i = K_d = 0$; run the feedforward trajectory; record target vs. measured position.
2. Identify position lag attributable to oil compressibility and friction.
3. Increase K_p to close the observed gap; if overshoot appears at trajectory peaks, increase K_d to damp it.
4. Introduce K_i only after K_p and K_d have been stabilised; verify anti-windup behaviour at the trajectory turning points before releasing the system for continuous operation.

8.5 Hardware Selection: Nepal Market

Table 3 compares the three most cost-competitive PLC platforms available in Nepal through South Asian distribution channels, ranked by approximate landed cost in Kathmandu. Prices are estimated from Indian market rates (IndiaMART, 2024–2025) with a $\times 1.5$ factor applied for Nepal customs duties, freight, and distributor margin.

Table 3: PLC hardware comparison for Nepal market, ranked by ascending cost

#	Platform	Est. NPR	Min. loop	Notes
1	**Coolmay EX2N**	14k–22k	3–5 ms	All-in-one CPU + 4.3" HMI + built-in 4AI/2AO. Cheapest entry point; instruction cycle $0.2\,\mu$s; practical closed-loop $\approx$ 300 Hz. Sourced via Indian distributors; lead time 1–2 weeks. Suitable when $f_{\text{op}} \lesssim 3$ Hz.
2	**Delta DVP28SV2**	26k–40k	2–4 ms	Separate high-speed CPU + expansion modules. Instruction cycle $0.24\,\mu$s; interrupt-based loop achieves $\approx$ 400–500 Hz. Strong motion-control set; broad local distributor network. Suitable for $f_{\text{op}} \lesssim 5$ Hz with tighter tracking.
3	**Mitsubishi FX5U**	48k–65k	0.5–1 ms	Approaches the 1 kHz spec. CPU instruction cycle 34 ns; FX5-4AD high-speed module converts at $125\,\mu$s/channel, enabling $\approx$ 750 Hz closed loop. Well-stocked at Kathmandu dealers. Required if $f_{\text{op}} > 5$ Hz or precision < 1 mm.

1 kHz rate note. Only the Mitsubishi FX5U with FX5-4AD modules can reliably approach 1 kHz closed-loop control. The Coolmay and Delta options are limited to $\approx$ 300–500 Hz by their standard analog module conversion times. However, given that the valve spool bandwidth is $\approx$ 50–100 Hz and the practical trajectory frequency ceiling is $\lesssim 20$ Hz (Section 7), a 500 Hz update rate from the Delta option provides a $\times 25$ sampling ratio above the trajectory frequency, which is more than sufficient for stable outer-loop PID control. The 1 kHz specification should therefore be confirmed as a genuine requirement—not a default—before selecting hardware.

9 I/O Architecture

Table 4 summarises the minimum process I/O required to implement the control strategy described in this proposal.

Table 4: Minimum I/O architecture for PLC implementation

I/O Type	**Min.**	**Connected Component**	**Function**
Analog Input (AI)	2	Piston position sensor, DCV spool sensor	Read $S_{act}(t)$ and $u_{dcv,actual}$ for PID feedback
Analog Output (AO)	2	DCV command, PCV command	Write normalised $u_{dcv} \in [-1, +1]$ and $u_{pcv} \in [0, +1]$
Digital Output (DO)	1	DCV enable pin	Hardware safety interlock
Digital Input (DI)	2	Start button, emergency stop	Operator control and safety

The normalised commands map to physical output signals as follows:

$$V_{\mathrm{DCV}} = 5 + 5\,u_{\mathrm{dcv}} \quad [\mathrm{V}] \qquad (0\,\mathrm{V} = \text{full retract; } 5\,\mathrm{V} = \text{centre; } 10\,\mathrm{V} = \text{full extend})$$
$$I_{\mathrm{DCV}} = 12 + 8\,u_{\mathrm{dcv}} \quad [\mathrm{mA}] \qquad (4\,\mathrm{mA} = \text{full retract; } 12\,\mathrm{mA} = \text{centre; } 20\,\mathrm{mA} = \text{full extend})$$
$$V_{\mathrm{PCV}} = 10\,u_{\mathrm{pcv}} \quad [\mathrm{V}] \qquad (0\,\mathrm{V} = \text{zero pressure; } 10\,\mathrm{V} = \text{full pump pressure})$$
$$I_{\mathrm{PCV}} = 4 + 16\,u_{\mathrm{pcv}} \quad [\mathrm{mA}] \qquad (4\,\mathrm{mA} = \text{zero; } 20\,\mathrm{mA} = \text{full pump pressure})$$

10 Summary and Recommendations

1. **PCV hardware type:** (Configuration A vs. B). Configuration A requires only the constant of Equation 4 for the PCV and fixed amplitude coefficients k_{ext}, k_{ret} (Equations 5–6) for the DCV; Configuration B requires real-time force tracking and a floating-point square-root evaluation every control cycle.

2. **Normalised command signal convention:** $u_{\text{dcv}} \in [-1, +1]$ because the DCV is bidirectional—negative values command retraction, positive command extension. $u_{\text{pcv}} \in [0, +1]$ is unidirectional. Both map linearly onto 0–10 V or 4–20 mA hardware interfaces per the equations in Section 9. If $|u_{\text{dcv,total}}|$ exceeds unity, the trajectory velocity or frequency must be reduced, or a higher-rated valve or pump must be specified.

3. **Evaluate the four frequency constraints** (Equation 30) before specifying the operating frequency. Both the hydraulic natural frequency f_h and the force-limited frequency f_F scale as $1/\sqrt{m}$: heavier loads permit significantly lower maximum frequencies. For medium-duty applications, the valve bandwidth (Equation 29) typically dominates, capping achievable tracking at 10–20 Hz regardless of load.

4. **Select PLC hardware against the actual required loop rate.** For trajectory frequencies $\lesssim 3$ Hz with moderate tolerance, the Coolmay EX2N-43H-24MT provides the lowest entry cost with an integrated HMI. For frequencies up to ≈ 5 Hz or tighter tolerance, the Delta DVP28SV2 combination is more capable. Only the Mitsubishi FX5U with FX5-4AD modules can approach the 1 kHz specification; this platform is warranted if tight accuracy or high-frequency operation is genuinely required.

5. **Include anti-windup and deadband compensation** in the PID implementation from the outset, given the end-stop risk in any hydraulic actuator during direction reversal.

6. **Characterise K_v against fluid temperature** across the 30–55 °C VG46 operating range; viscosity-driven variation in K_v directly affects the accuracy of the feedforward flow calculations.

7. **Commission in stages**: validate open-loop feedforward first, then introduce K_p, K_d, and K_i incrementally per Section 8.4.